In Llama, Saying "I Don't Know" Tracks Entity Popularity More Than Knowledge

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Abstract

When a language model answers a factual question with "I don't know", is it reporting uncertainty about the answer? We cross entity popularity in PopQA with knowledge labels taken from the model's own sampled answers. In Llama-3.1-8B-Instruct, abstention follows popularity more than knowledge: about obscure entities the model says "I don't know" in 0.43 of replies even for facts it answers correctly when made to answer, against 0.00 for popular entities. This holds with knowledge relabelled from a different prompt, with a third judge, without guessable answers, and controlling for what the model knows about the entity. First-token answer entropy instead follows knowledge, in four model families and under the independent labels; within one prompt, the first token's lean towards "I don't know" follows popularity while its entropy follows knowledge. Mechanistic tests narrow the picture: steering along an obscure-minus-popular direction adds little abstention, and only through its abstention component; base models' decoded hedge readout responds to popularity but as much to knowledge. Gemma-2-9B, the only other family whose abstention we could measure, shows no popularity preference.

1 Introduction

A model that says "I don't know" when it lacks a fact avoids the worst hallucinations [Kalai et al., 2025]. Confidence estimates usually read uncertainty over answers [Kadavath et al., 2022, Kuhn et al., 2023, Stolfo et al., 2024], and models use such confidence when the prompt invites them to abstain [Kumaran et al., 2026]. Verbal and answer-level uncertainty are known to diverge [Ji et al., 2025]; what each follows, and whether different models follow the same variable, is less clear.

People's sense that they can answer a question follows how familiar its terms are, not whether the answer is stored [Reder and Ritter, 1992, Schwartz and Metcalfe, 1992]. In language models, "known entity" features suppress a default "can't answer" response, and Lindsey et al. [2025] propose that a familiar name can trigger them without the fact. Tests of this idea define an entity as known when the model answers about it correctly [Ferrando et al., 2025], so familiarity and knowledge move together.

We separate them. From PopQA [Mallen et al., 2023] we take questions about obscure and popular entities and label each known or unknown from the model's own answers [Gekhman et al., 2024]. The deciding observation is the cell where the two disagree: facts about obscure entities that the model answers correctly every time it is made to answer. Our contribution is this factorial test of the misfire idea, its robustness checks, and a comparison across families

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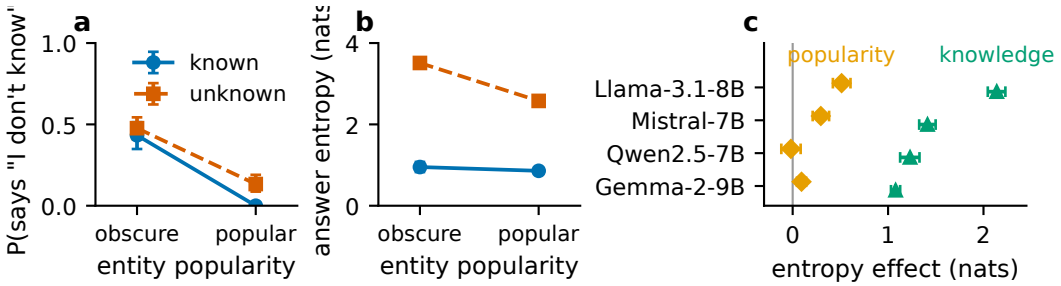


Figure 1: **In Llama, saying "I don't know" follows popularity, while answer uncertainty follows knowledge.** (a) Llama-3.1-8B-Instruct: share of free replies both judges label "I don't know" ($n = 387$ questions, 3 replies each; obscure known $n = 87$). (b) Its first-token entropy under the forced-answer prompt ($n = 2{,}303$). (c) Popularity and knowledge effects on that entropy in four families. Means with 95% item-bootstrap intervals.

of what abstention and answer uncertainty follow (Sections 3.1 and 3.2); steering and lens analyses then narrow the mechanistic reading (Section 3.3).

2 Setup

Questions and labels. We use the 16 relations of PopQA. Within each relation, subjects in the bottom and top deciles of Wikipedia page views form the *obscure* and *popular* arms; page-view popularity is a proxy for familiarity, and we report it as popularity. For each model, a greedy answer and ten samples ($T = 0.7$) under a *forced-answer prompt* (the chat template plus the assistant prefix "The answer is") are graded against gold aliases: a fact is *known* if the greedy answer and at least 90% of samples are correct and *unknown* if none is; the rest are dropped. This leaves 2,303, 2,392, 2,282 and 2,378 questions for Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, Gemma-2-9B-it and Mistral-7B-Instruct-v0.3 [Grattafiori et al., 2024, Qwen et al., 2024, Team et al., 2024, Jiang et al., 2023].

Two readouts. *Answer uncertainty* is the first-token entropy under the forced-answer prompt, whose next token starts an answer. Because the knowledge labels are sampled from the same distribution, entropy is expected to follow knowledge, so we also relabel knowledge from an independent prompt (Section 3.1). *Abstention* needs the option to decline, so it is measured with a *free prompt*, the chat template alone: for up to 100 questions per popularity $\times$ knowledge cell (387 in total) we draw three replies ($T = 0.7$, up to 96 tokens). Two LLM judges (Qwen2.5-32B-Instruct, Gemma-2-27B-it) label each reply DONT_KNOW, CLARIFY (asks which entity is meant) or ANSWER, and a reply abstains when both say DONT_KNOW. On Llama the judges agree at $\kappa = 0.62$, above a pre-set bar of 0.60; on the other families they do not (Section 3.2). No human validated the judges.

Effects. The popularity effect is the obscure-minus-popular difference in a readout averaged over the two knowledge levels, and the knowledge effect the unknown-minus-known difference averaged over popularity, with 95% item-bootstrap intervals. Confirmatory analyses followed criterion notes hashed before each run (Appendix H); the rest are post hoc.

3 Results

3.1 In Llama, abstention follows popularity; answer uncertainty follows knowledge

About obscure entities, Llama says "I don't know" in 0.43 of replies to questions it answers correctly when made to answer, against 0.00 about popular entities (Figure 1a), while forced-answer entropy in these two cells is nearly equal (Figure 1b). Over the whole design the popularity effect on abstention is +0.39 [+0.33, +0.45] and the knowledge effect +0.09 [+0.03, +0.15]; their difference, +0.30 [+0.20, +0.40], keeps its sign under four other measurements,

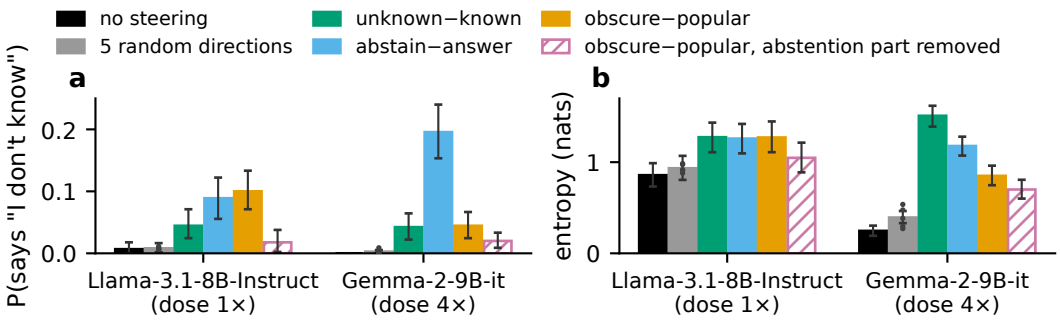


Figure 2: **Steering along the obscure-minus-popular direction adds little abstention beyond a generic abstention direction.** (a) Share of free replies both judges label "I don't know" on 150 popular questions each model knows, held out from fitting and from choosing the dose; dots: individual random directions. (b) First-token entropy under the forced-answer prompt. Error bars: 95% bootstrap over questions.

which range from +0.13 to +0.44 (Appendix A). Answer entropy shows the reverse: it rises by +2.14 nats when Llama lacks the fact and by +0.51 when the entity is obscure, and in all four families the knowledge effect is at least four times the popularity effect (Figure 1c).

Five checks, fixed in advance unless stated, address the labels and the measurement. *Independent labels:* relabelling knowledge with a paraphrased question and a different prefix, the popularity effect on abstention is +0.38 against +0.06 for knowledge (difference +0.32 [+0.16, +0.49]; obscure known 0.46, $n = 27$), and entropy still follows knowledge (knowledge minus popularity +1.40 [+1.02, +1.73]). *Third judge:* adding Llama-3.3-70B and counting a reply as abstaining when two of three judges agree raises agreement to $\kappa = 0.74$ and gives a difference of +0.44 [+0.32, +0.55]. *Entity knowledge:* Llama correctly says what the entity is for 0.83 of obscure known facts (popular: 0.75); this entity-level knowledge tracks page views (ρ =+0.52) yet, as a covariate, shrinks the popularity effect by only 5% (+0.34 [+0.27, +0.41]). *Guessable answers* (post hoc; Mallen et al., 2023): without them the difference stays +0.30 [+0.19, +0.41]. *One prompt* (descriptive): at the free prompt itself, the log-odds that the first token is "I" follow popularity (+8.4 against +3.1) while its entropy follows knowledge (+0.16 against +0.05), so the split does not depend on comparing two prompts. The pattern matches the cue-familiarity account of human feelings of knowing [Reder and Ritter, 1992, Schwartz and Metcalfe, 1992], though page views are not the model's own familiarity: name surprisal under the model did not even correlate with them (Appendix F).

3.2 Across families: shared answer uncertainty, different abstention

The entropy pattern recurs in all four families (Figure 1c); abstention does not. Qwen, Gemma and Mistral rarely say "I don't know" and the two judges disagree on them; with a third judge, Gemma reaches $\kappa = 0.64$ but its abstention shows no preference for popularity over knowledge (difference +0.01 [−0.03, +0.06]; at most 0.11 even for obscure, unknown facts), while Qwen ($\kappa = 0.41$) and Mistral ($\kappa = 0.54$) remain unmeasured (Appendix B). The popularity preference is thus Llama's, not general.

3.3 Mechanistic tests narrow the reading

Steering. Earlier work steers refusal along recognition directions [Ferrando et al., 2025, Brzezinka, 2026]. We fitted obscure-minus-popular, unknown-minus-known and abstain-minus-answer directions on half of the questions [Arditi et al., 2024, Rimsky et al., 2024], chose the dose on a second set and scored a third with both judges (Appendix C). The obscure-minus-popular direction raised Llama's abstention from 0.01 to 0.10 (+0.09 [+0.06, +0.12] over random directions, short of the 0.10 bar fixed in advance; +0.17 with a majority of three judges) and Gemma's from 0.00 to 0.04 at 4× its natural norm (Figure 2a). In Llama this direction is nearly parallel to the abstention direction (cosine 0.95), and with that component removed it adds +0.01 [−0.00, +0.02] (+0.02 with three judges; Gemma

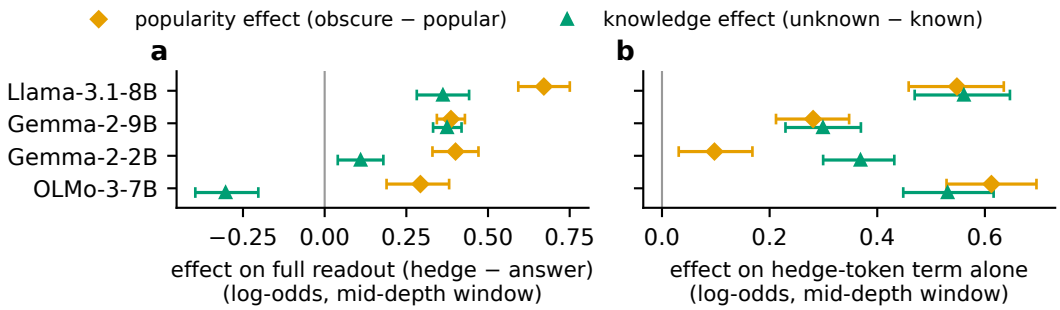


Figure 3: **In base models, the decoded hedge readout responds to popularity, and about as much to knowledge.** Jacobian-lens readout at the last position of "Question: ... Answer: The answer is", averaged over the pre-set mid-depth window, for four base models: (a) log-odds of hedge tokens against the answer token; (b) the hedge-token term alone. Points: popularity and knowledge effects with 95% item-bootstrap intervals. Lenses are fitted per model, so magnitudes are not comparable across models.

in Appendix C). Steering thus moves a generic abstention signal rather than isolating a familiarity cause [cf. Che and Li, 2026]. Only entropy separates the directions, and only in Gemma: unknown-minus-known raises it +0.66 [+0.58, +0.74] nats more (Figure 2b).

Base models. The Jacobian lens maps a layer's activations to scores over output tokens [Gurnee et al., 2026]; with its prefitted lenses we read a fixed set of hedge tokens against the answer token in four base models of three families, at a forced-answer position. In the mid-depth window fixed before the run, the obscure-minus-popular effect is positive in all four models, and it sits in the hedge-token term itself rather than only in how rare the answer token is (Figure 3b; all four intervals exclude zero). But the knowledge effect on that term is about as large (Figure 3b), and on the full readout OLMo's knowledge effect is negative (Figure 3a; Appendix D). A popularity-sensitive hedge signal thus exists before instruction tuning, while Llama-Instruct's preference for popularity over knowledge is not seen in base models. The human-memory prediction that familiarity acts at a shallower depth than knowledge [Koriat and Levy-Sadot, 2001] failed (1 supporting, 1 contradicting, 6 undecided of eight models; Appendix D).

4 Related work

Entity-recognition features that gate refusal carry over from base to chat models [Ferrando et al., 2025]; Lindsey et al. [2025] describe the misfire hypothesis we test. Brzezinka [2026] separates graded familiarity probes from the abstention policy, without a knowledge axis. Models threshold internal confidence when abstention is offered in a multiple-choice prompt [Kumaran et al., 2026]; ours is uninstructed and free-form. Confidence also tracks how popular an answer is [Ni et al., 2025], and verbal and semantic uncertainty diverge [Ji et al., 2025]; we add what each follows, and that it differs across families.

5 Limitations and conclusion

Abstention is judged by LLMs not validated against human labels, and the popularity preference is shown for one model; Gemma lacks it. Popularity is a page-view proxy, models have 2–9B parameters, and questions are single-entity PopQA questions. Neither steering nor the lens isolates a familiarity cause.

In Llama, answer uncertainty follows knowledge and abstention follows popularity, so predicting abstention from answer uncertainty misses its main driver; calibration work should measure both. That only the first holds in every family suggests the two are shaped separately.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems 37*, NeurIPS 2024, page 136037–136083. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024. doi: 10.52202/079017-4322. URL http://dx.doi.org/10.52202/079017-4322.

Grzegorz Brzezinka. Graded entity-familiarity readouts in language models: Polish adaptation, cross-language robustness, and refusal steering, 2026. URL https://arxiv.org/abs/2607.13568.

Tong Che and Yilong Li. Does the model use the feature? separating steering from mechanism in llms, 2026. URL https://arxiv.org/abs/2610.07270.

Javier Ferrando, Oscar Obeso, Senthooran Rajamanoharan, and Neel Nanda. Do i know this entity? knowledge awareness and hallucinations in language models, 2025. URL https://openreview.net/forum?id=WCRQFlji2q.

Zorik Gekhman, Gal Yona, Roee Aharoni, Matan Eyal, Amir Feder, Roi Reichart, and Jonathan Herzig. Does fine-tuning llms on new knowledge encourage hallucinations? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, page 7765–7784. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-main.444. URL http://dx.doi.org/10.18653/v1/2024.emnlp-main.444.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yeary, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas

Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti, Vítor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuewei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkang Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippos Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabsa, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will

Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models, 2024. URL https://arxiv.org/abs/2407.21783.

Wes Gurnee, Theo Horsley, Zifan Carl Guo, Tara Rezaei Kheirkhah, Qinyi Sun, Will Hathaway, Neel Nanda, and Dimitris Bertsimas. Universal neurons in gpt2 language models, 2024. URL https://openreview.net/forum?id=ZeI104QZ8I.

Wes Gurnee, Nicholas Sofroniew, Adam Pearce, Mateusz Piotrowski, Isaac Kauvar, Runjin Chen, Anna Soligo, Paul Bogdan, Euan Ong, Rowan Wang, Ben Thompson, David Abrahams, Subhash Kantamneni, Emmanuel Ameisen, Joshua Batson, and Jack Lindsey. Verbalizable representations form a global workspace in language models, 2026. URL https://arxiv.org/abs/2607.15495.

Ziwei Ji, Lei Yu, Yeskendir Koishekenov, Yejin Bang, Anthony Hartshorn, Alan Schelten, Cheng Zhang, Pascale Fung, and Nicola Cancedda. Calibrating verbal uncertainty as a linear feature to reduce hallucinations. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, page 3769–3793. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.187. URL http://dx.doi.org/10.18653/v1/2025.emnlp-main.187.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7b, 2023. URL https://arxiv.org/abs/2310.06825.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know, 2022. URL https://arxiv.org/abs/2207.05221.

Adam Tauman Kalai, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. Why language models hallucinate, 2025. URL https://arxiv.org/abs/2509.04664.

Asher Koriat and Ravit Levy-Sadot. The combined contributions of the cue-familiarity and accessibility heuristics to feelings of knowing. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 27(1):34–53, 2001. ISSN 0278-7393. doi: 10.1037/0278-7393.27.1.34. URL http://dx.doi.org/10.1037/0278-7393.27.1.34.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation, 2023. URL https://arxiv.org/abs/2302.09664.

Dharshan Kumaran, Nathaniel Daw, Simon Osindero, Petar Veličković, and Viorica Patraucean. Causal Evidence that Language Models use Confidence to Drive Behavior, 2026. URL https://arxiv.org/abs/2603.22161.

Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the Biology of a Large Language Model. Transformer Circuits Thread, 2025. URL https://transformer-circuits.pub/2025/attribution-graphs/biology.html.

Yiqi Liu, Yang Wang, Songxin Wang, Chenghao Xiao, and Chenghua Lin. Broken symmetry in llm refusal: Answer release is more local than refusal restoration, 2026. URL https://arxiv.org/abs/2608.15772.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 9802–9822. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.acl-long.546. URL http://dx.doi.org/10.18653/v1/2023.acl-long.546.

Vy Nguyen, Ziqi Xu, Jeffrey Chan, Estrid He, Feng Xia, Renqiang Luo, Erik Cambria, and Xiuzhen Zhang. The commit-abstain circuit: Why language models hallucinate instead of abstaining, 2026. URL https://arxiv.org/abs/2609.32964.

Shiyu Ni, Keping Bi, Jiafeng Guo, and Xueqi Cheng. Popular but Wrong: Understanding and Mitigating LLM Overconfidence through Knowledge Popularity, 2025. URL https://arxiv.org/abs/2505.17537. EMNLP 2026 (main conference).

Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2024. URL https://arxiv.org/abs/2412.15115.

Lynne M. Reder and Frank E. Ritter. What determines initial feeling of knowing? familiarity with question terms, not with the answer. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 18(3):435–451, May 1992. ISSN 0278-7393. doi: 10.1037/0278-7393.18.3.435. URL http://dx.doi.org/10.1037/0278-7393.18.3.435.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 15504–15522. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.828. URL http://dx.doi.org/10.18653/v1/2024.acl-long.828.

Bennett L. Schwartz and Janet Metcalfe. Cue familiarity but not target retrievability enhances feeling-of-knowing judgments. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 18(5):1074–1083, 1992. ISSN 0278-7393. doi: 10.1037/0278-7393.18.5.1074. URL http://dx.doi.org/10.1037/0278-7393.18.5.1074.

Alessandro Stolfo, Ben Wu, Wes Gurnee, Yonatan Belinkov, Xingyi Song, Mrinmaya Sachan, and Neel Nanda. Confidence regulation neurons in language models. In *Advances in Neural Information Processing Systems 37*, NeurIPS 2024, page 125019–125049. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024. doi: 10.52202/079017-3970. URL http://dx.doi.org/10.52202/079017-3970.

Aaquib Syed, Can Rager, and Arthur Conmy. Attribution patching outperforms automated circuit discovery. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, page 407–416. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.blackboxnlp-1.25. URL http://dx.doi.org/10.18653/v1/2024.blackboxnlp-1.25.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, Johan Ferret, Peter Liu, Pouya Tafti, Abe Friesen, Michelle Casbon, Sabela Ramos, Ravin Kumar, Charline Le Lan, Sammy Jerome, Anton Tsitsulin, Nino Vieillard, Piotr Stanczyk, Sertan Girgin, Nikola Momchev, Matt Hoffman, Shantanu Thakoor, Jean-Bastien Grill, Behnam Neyshabur, Olivier Bachem, Alanna Walton, Aliaksei Severyn, Alicia Parrish, Aliya Ahmad, Allen Hutchison, Alvin Abdagic, Amanda Carl, Amy Shen, Andy Brock,

Andy Coenen, Anthony Laforge, Antonia Paterson, Ben Bastian, Bilal Piot, Bo Wu, Brandon Royal, Charlie Chen, Chintu Kumar, Chris Perry, Chris Welty, Christopher A. Choquette-Choo, Danila Sinopalnikov, David Weinberger, Dimple Vijaykumar, Dominika Rogozińska, Dustin Herbison, Elisa Bandy, Emma Wang, Eric Noland, Erica Moreira, Evan Senter, Evgenii Eltyshev, Francesco Visin, Gabriel Rasskin, Gary Wei, Glenn Cameron, Gus Martins, Hadi Hashemi, Hanna Klimczak-Plucińska, Harleen Batra, Harsh Dhand, Ivan Nardini, Jacinda Mein, Jack Zhou, James Svensson, Jeff Stanway, Jetha Chan, Jin Peng Zhou, Joana Carrasqueira, Joana Iljazi, Jocelyn Becker, Joe Fernandez, Joost van Amersfoort, Josh Gordon, Josh Lipschultz, Josh Newlan, Ju-yeong Ji, Kareem Mohamed, Kartikeya Badola, Kat Black, Katie Millican, Keelin McDonell, Kelvin Nguyen, Kiranbir Sodhia, Kish Greene, Lars Lowe Sjoesund, Lauren Usui, Laurent Sifre, Lena Heuermann, Leticia Lago, Lilly McNealus, Livio Baldini Soares, Logan Kilpatrick, Lucas Dixon, Luciano Martins, Machel Reid, Manvinder Singh, Mark Iverson, Martin Görner, Mat Velloso, Mateo Wirth, Matt Davidow, Matt Miller, Matthew Rahtz, Matthew Watson, Meg Risdal, Mehran Kazemi, Michael Moynihan, Ming Zhang, Minsuk Kahng, Minwoo Park, Mofi Rahman, Mohit Khatwani, Natalie Dao, Nenshad Bardoliwalla, Nesh Devanathan, Neta Dumai, Nilay Chauhan, Oscar Wahltinez, Pankil Botarda, Parker Barnes, Paul Barham, Paul Michel, Pengchong Jin, Petko Georgiev, Phil Culliton, Pradeep Kuppala, Ramona Comanescu, Ramona Merhej, Reena Jana, Reza Ardeshir Rokni, Rishabh Agarwal, Ryan Mullins, Samaneh Saadat, Sara Mc Carthy, Sarah Cogan, Sarah Perrin, Sébastien M. R. Arnold, Sebastian Krause, Shengyang Dai, Shruti Garg, Shruti Sheth, Sue Ronstrom, Susan Chan, Timothy Jordan, Ting Yu, Tom Eccles, Tom Hennigan, Tomas Kocisky, Tulsee Doshi, Vihan Jain, Vikas Yadav, Vilobh Meshram, Vishal Dharmadhikari, Warren Barkley, Wei Wei, Wenming Ye, Woohyun Han, Woosuk Kwon, Xiang Xu, Zhe Shen, Zhitao Gong, Zichuan Wei, Victor Cotruta, Phoebe Kirk, Anand Rao, Minh Giang, Ludovic Peran, Tris Warkentin, Eli Collins, Joelle Barral, Zoubin Ghahramani, Raia Hadsell, D. Sculley, Jeanine Banks, Anca Dragan, Slav Petrov, Oriol Vinyals, Jeff Dean, Demis Hassabis, Koray Kavukcuoglu, Clement Farabet, Elena Buchatskaya, Sebastian Borgeaud, Noah Fiedel, Armand Joulin, Kathleen Kenealy, Robert Dadashi, and Alek Andreev. Gemma 2: Improving open language models at a practical size, 2024. URL https://arxiv.org/abs/2408.00118.

Carter Teplica, Yixin Liu, Arman Cohan, and Tim G. J. Rudner. SCIURus: Shared circuits for interpretable uncertainty representations in language models. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 12451–12469, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.618. URL https://aclanthology.org/2025.naacl-long.618/.

Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small, 2023. URL https://openreview.net/forum?id=XBflsCjpUM.

A Robustness of the Llama abstention result

The single corrected judge gives the smallest difference. Its 24-token replies are often cut off before the model commits; the two-judge measurement with 96 tokens was designed for that reason and made primary before it ran. The regex detector's recall on the 96-token replies, against the two-judge labels, is 0.99.

Independent labels (criterion hashed before the run). Knowledge was relabelled with one paraphrased question per relation, the assistant prefix "Answer:" and fresh samples, under the same 90%/0% rule. Agreement with the original labels is moderate, mainly because the paraphrase moves many popular known facts to the intermediate band, which is dropped. Under the new labels, the popularity effect on abstention is +0.38 [+0.29, +0.47] and the knowledge effect +0.06 [−0.03, +0.15]; original-prompt entropy follows the new labels (knowledge minus popularity +1.40 [+1.02, +1.73]). At the free prompt, the first token's "I"

Table 1: Popularity effect minus knowledge effect on Llama's abstention under each measurement and labelling. Positive values mean abstention follows popularity more than knowledge. "Corrected": regex rates on 24-token replies corrected for the recall and precision of one LLM judge (Qwen2.5-32B-Instruct), measured on a subset labelled by that judge, not by people.

Measurement	Replies	Popularity − knowledge
Regex detector	20 per question, 24 tokens	+0.44 [+0.34, +0.53]
One LLM judge, corrected	same 24-token replies	+0.13 [+0.03, +0.24]
Two judges, both DONT_KNOW (primary)	3 per question, 96 tokens	+0.30 [+0.20, +0.40]
Two judges, DONT_KNOW or CLARIFY	same	+0.31 [+0.21, +0.41]
Regex detector	same	+0.43 [+0.31, +0.55]
Three judges, majority (third judge Llama-3.3-70B)	same	+0.44 [+0.32, +0.55]
Two judges, knowledge relabelled (paraphrased prompt)	same	+0.32 [+0.16, +0.49]
Two judges, guessable answers removed (post hoc)	same	+0.30 [+0.19, +0.41]
Two judges, entity knowledge as covariate (popularity coefficient)	same	+0.34 [+0.27, +0.41]

log-odds give popularity +8.4 against knowledge +3.1 (difference +5.3 [+4.4, +6.0]), and its entropy popularity +0.05 against knowledge +0.16 (difference −0.11 [−0.18, −0.03]).

Entity knowledge (criterion hashed before the run). Llama completed "{subject} is" after the question "What is {subject}?" (greedy and five samples), and Llama-3.3-70B graded each completion against the subject's English Wikidata description; entity knowledge is the share graded correct. It averages 0.83 for obscure known facts and 0.75 for popular known facts. In a linear probability model with relation fixed effects, its coefficient is −0.03 [−0.06, +0.00] per standard deviation, and it shrinks the popularity coefficient by 5% on the two-judge measure and 3% on the regex measure.

Guessable answers (post hoc). An answer counts as guessable if a gold alias of at least three characters appears in the subject's name, or if it is its relation's most common answer and covers at least 20% of the relation's pool. Among unknown facts alone the popularity gap in abstention is +0.34 [+0.26, +0.43], which mislabelled guesses cannot produce. On the regex measure over the 2,106 non-guessable questions the difference is +0.48 [+0.36, +0.59].

Free replies (post hoc). In the obscure known cell, 0.53 of free replies contain a gold alias, and 0.72 of the replies that do not abstain; for 24 of 87 questions, one free reply gives the correct answer and another says "I don't know". A linear probability model of abstention with relation fixed effects gives popularity +0.29 [+0.23, +0.36], knowledge −0.05 [−0.14, +0.03] and +0.14 per standard deviation of forced-answer entropy.

Name spelling (post hoc). Over the 2,303 labelled questions with relation fixed effects, the popularity coefficient on the regex measure moves from −1.87 to −1.79 logits per standard deviation after adding the name's word frequency, token count and token rarity. Among 55 matched pairs of known facts that differ in popularity, the abstention gap of +0.76 falls to +0.43 [+0.18, +0.69] at equal surface form.

B Abstention and the forced-answer hedge readout in all four families

Intervals for the popularity-minus-knowledge rows: three judges, Llama [+0.32, +0.55], Gemma [−0.03, +0.06]; hedge log-odds, Llama [−0.72, +0.45], Qwen [−3.41, −1.64], Gemma [+1.79, +2.42], Mistral [−0.55, +1.34]. The two judges that agree best across families are Qwen2.5-32B and Llama-3.3-70B ($\kappa = 0.96$ on Llama). At the forced-answer position,

Table 2: Judge agreement and abstention. Model-level rows: judge agreement and regex recall. Cell rows: abstention in the obscure, unknown cell. Bottom: effects on first-token hedge-versus-answer log-odds at the forced-answer position.

	Llama	Qwen	Gemma	Mistral
Two judges, κ (bar 0.60)	0.62	0.30	0.47	0.39
Three judges, Fleiss κ (bar 0.60)	0.74	0.41	0.64	0.54
Regex recall against two judges	0.99	0.58	0.48	0.73
Both judges DONT_KNOW	0.48	0.05	0.05	0.10
Both judges DONT_KNOW or CLARIFY	0.65	0.32	0.22	0.25
Three judges, majority	0.75	0.10	0.11	0.19
Regex detector	0.85	0.18	0.07	0.21
Three judges: popularity − knowledge	+0.44	–	+0.01	–
Hedge log-odds: popularity	+2.93	+1.14	+1.79	+2.48
Hedge log-odds: knowledge	+3.05	+3.63	−0.32	+2.08
Hedge log-odds: popularity − knowledge	−0.12	−2.49	+2.11	+0.40

Llama's hedge readout follows popularity and knowledge about equally, which is why the one-prompt check in Section 3.1 uses the free prompt.

C Steering details

Criterion hashed before the run (Appendix H). Directions were fitted on a random half of each model's questions at layer round$(0.55 \times (L-1))$, the last position of the free prompt: obscure minus popular and unknown minus known (differences of cell means, each averaged over the other factor), and abstain minus answer (items whose stored free-generation regex abstention rate is at least 0.5 minus items with rate 0). The obscure-minus-popular direction with its abstention component projected out was rescaled to the original norm. Each direction was added at a multiple $k \in \{0.5, 1, 2, 4, 8\}$ of the obscure-minus-popular direction's natural norm, at the last prompt position and every generated position. k was chosen on 150 popular known questions as the smallest value at which that direction beat five random directions by 0.10 in regex abstention (1× for Llama, 4× for Gemma), and all conditions were scored on 150 further popular known questions with both judges.

Table 3: Share of steered replies both judges label "I don't know" (regex detector in parentheses), 150 evaluation questions, 3 replies each.

	None	Random	Unk.−known	Abstain−answer	Obscure−popular	... abstention removed
Llama	0.01 (0.03)	0.008	0.04 (0.10)	0.09 (0.25)	0.10 (0.23)	0.02 (0.02)
Gemma	0.00 (0.00)	0.003	0.04 (0.03)	0.20 (0.74)	0.04 (0.27)	0.02 (0.07)

Obscure-minus-popular minus random: Llama +0.09 [+0.06, +0.12], Gemma +0.04 [+0.02, +0.06] (secondary scorer, a majority of three judges including Llama-3.3-70B: +0.17 [+0.13, +0.22] and +0.11 [+0.08, +0.15]; with abstention removed +0.02 [+0.00, +0.04] and +0.05 [+0.03, +0.07]); with abstention removed: +0.01 [−0.00, +0.02] and +0.02 [+0.01, +0.03]; obscure-minus-popular minus unknown-minus-known: +0.06 [+0.02, +0.09] and +0.00 [−0.03, +0.03]. The regex detector counts far more abstentions than the judges for steered replies, so an earlier, regex-scored version of this experiment with the dose chosen on the evaluation questions (Llama 0.02 to 0.29) overstated the effect. The cosine between the obscure-minus-popular and abstention directions is 0.95 in Llama and 0.63 in Gemma; since abstaining items are mostly obscure, part of this overlap is built in. No condition produced degenerate text.

D Lens: window effects, onsets and the instruct models

Criterion hashed before the run. Hedge term: log-sum-exp of the lens log-probabilities of a fixed set of hedge tokens; the full readout subtracts the same quantity for the answer tokens.

Window: relative depth $\ell/(L-1) \in [1/3, 2/3]$. Intervals: 2,000 within-cell item bootstraps. Base models were labelled known or unknown on their own prompt with the same sampling rule.

Table 4: Mid-depth window effects in base models, with 95% intervals.

	Full readout		Hedge term alone	
Model	Popularity	Knowledge	Popularity	Knowledge
Llama-3.1-8B	+0.67 [+0.59, +0.75]	+0.36 [+0.28, +0.44]	+0.55 [+0.46, +0.64]	+0.56 [+0.47, +0.65]
Gemma-2-9B	+0.39 [+0.34, +0.43]	+0.37 [+0.33, +0.42]	+0.28 [+0.21, +0.35]	+0.30 [+0.23, +0.37]
Gemma-2-2B	+0.40 [+0.33, +0.47]	+0.11 [+0.04, +0.18]	+0.10 [+0.03, +0.17]	+0.37 [+0.30, +0.43]
OLMo-3-7B	+0.29 [+0.19, +0.38]	−0.30 [−0.40, −0.20]	+0.61 [+0.53, +0.70]	+0.53 [+0.45, +0.62]

The onset of an effect curve is the first layer where it reaches half its positive maximum over layers and its 95% interval excludes zero. A model supports the shallower-familiarity prediction if the knowledge onset follows the popularity onset with an interval above zero, and contradicts it if the interval lies below zero.

Table 5: Lens onsets (layer index) of the popularity (F) and knowledge (K) effects, their difference $D = K - F$, the outcome, and the mid-depth popularity effect on the full readout, for base and instruct models.

Model	Layers	F onset	K onset	D	Outcome	Mid popularity
Llama-3.1-8B-Instruct	32	17	17	0	undecided	+1.58
Llama-3.1-8B	32	16	16	0	undecided	+0.67
Qwen2.5-7B-Instruct	28	20	20	0	undecided	+0.21
Gemma-2-9B-it	42	24	24	0	undecided	+0.71
Gemma-2-9B	42	24	25	+1	supports	+0.39
Gemma-2-2B-it	26	17	15	-2	undecided	+0.40
Gemma-2-2B	26	16	16	0	undecided	+0.40
OLMo-3-7B	32	19	14	-5	contradicts	+0.29

E Cross-model agreement (exploratory)

Spearman correlations across questions between Llama's and each other model's forced-answer hedge log-odds and first-token entropies, on questions where both models have the same knowledge label and where they differ (95% item-bootstrap intervals). The plan, including the rule that hedge agreement and the hedge-minus-entropy gap both have 95% intervals excluding zero, was written before this run, but the point estimates had been seen, so the analysis is post hoc.

	Llama–Qwen	Llama–Gemma	Llama–Mistral
Questions, knowledge differs	313	60	108
Hedge log-odds agreement	+0.11 [−0.00, +0.22]	+0.38 [+0.13, +0.60]	+0.25 [+0.06, +0.42]
Entropy agreement	−0.17 [−0.28, −0.07]	+0.03 [−0.24, +0.28]	−0.07 [−0.27, +0.12]
Gap (hedge − entropy)	+0.28 [+0.14, +0.42]	+0.35 [−0.03, +0.74]	+0.32 [+0.06, +0.57]
Rule	not met	not met	met
Questions, same knowledge	1,784	1,977	2,000
Hedge log-odds agreement	+0.50	+0.38	+0.64
Entropy agreement	+0.46	+0.54	+0.69

F A model-side familiarity measure (negative result)

Criterion hashed before the run. We computed the mean per-token surprisal of each subject name under the base Llama-3.1-8B in the context "Here is a name:". It is uncorrelated with log page views (Spearman −0.001), and with knowledge and name length held fixed its coefficient on two-judge abstention is +0.013 [−0.026, +0.059] per standard deviation, so

it neither stands in for popularity nor predicts abstention. Short names made of common words ("New York" as a book title) are a likely reason.

G Timing, neurons and the circuit

On matched pairs of obscure and popular questions in Llama, the lens-decoded abstention readout reaches half its final size at layer 16, while the entropy gap does so at layer 28, 12 layers later; this compares two readouts of one contrast, not the onsets in Appendix D. Neurons selected for their correlation with output entropy in Llama and Qwen behaved as word-frequency and temperature neurons [Stolfo et al., 2024, Gurnee et al., 2024]. A set of 20 attention heads and 100 MLP neurons found by activation patching [Wang et al., 2023, Syed et al., 2024] reproduces 0.83 [0.78, 0.87] of the first-token abstention log-odds difference between matched pairs; in generation, removing its state removes abstention in 0.98 of cases but injecting it produces abstention in only 0.40, short of a behavioural bar fixed in advance. Shared uncertainty circuits [Teplica et al., 2025], a commit-or-abstain circuit across ten models [Nguyen et al., 2026] and a similar asymmetry between releasing and restoring refusal [Liu et al., 2026] have been reported before.

H Analysis timing

Each confirmatory analysis had a criterion note written before its run, stating the measurement, the test and the wording each outcome would license; the SHA-256 prefix of each note was recorded before the run. Four-family design: eba35dec, b53d9a27; two-judge measurement: 81acfc9f; lens study: c8e04372; circuit behaviour: 1c99afd4; independent labels: af50f53f; name surprisal: 2b69e06c; lens split: b2b916ce; steering with held-out dose: 18c74883 (amendment adding the third judge as a secondary scorer: 7bed3826); third judge: f35aa4d9; entity knowledge: 937b46e3. An earlier steering experiment (da0ad000, 797f52d4) is superseded by the held-out design. The guessability, spelling, free-reply and entropy-control checks and the cross-model agreement were decided after seeing data and are post hoc.